

EXPERIMENT-1

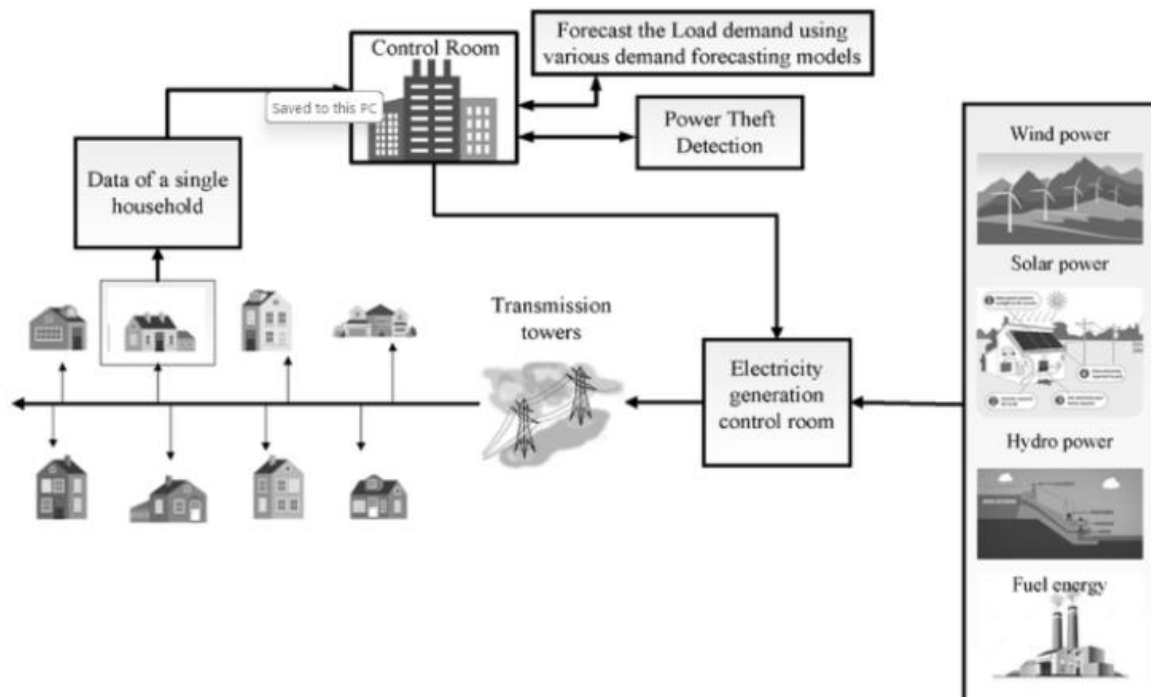
CODE:

```
import cv2

from google.colab.patches import cv2_imshow

img = cv2.imread('/content/Screenshot 2026-05-06 061946.png')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
cv2_imshow(gray)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

output:

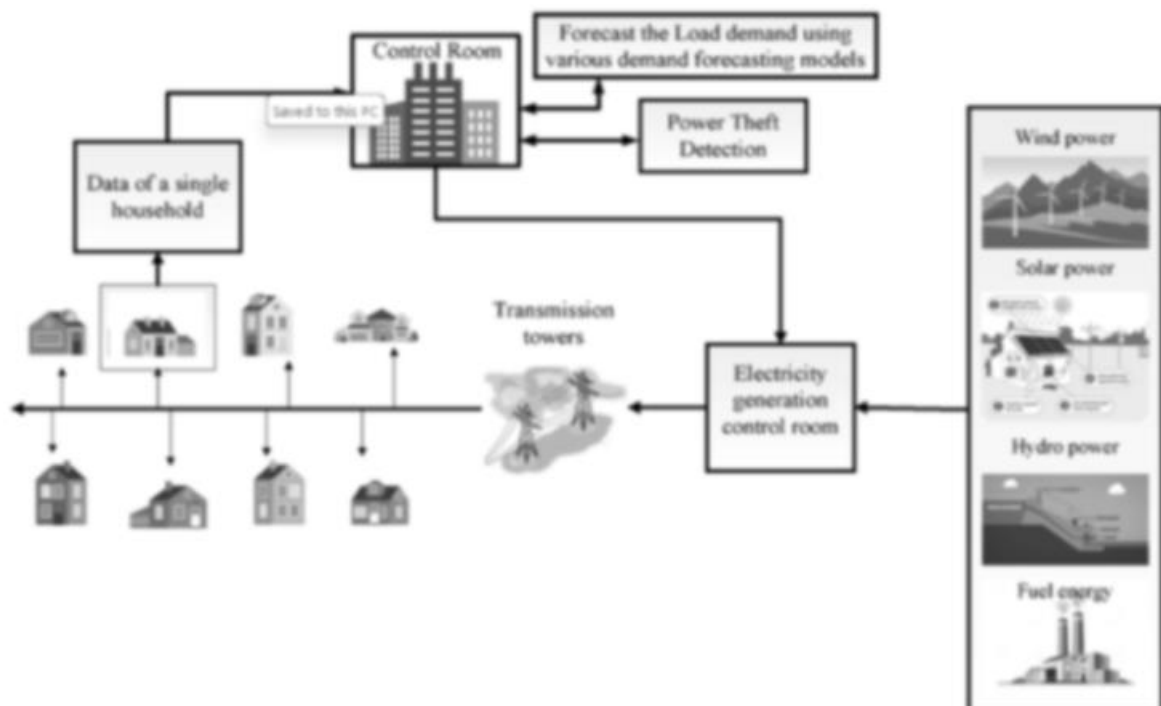


EXPERIMENT-2

CODE

```
import cv2  
  
from google.colab.patches import cv2_imshow  
  
img = cv2.imread('/content/Screenshot 2026-05-06 061946.png')  
  
blur = cv2.GaussianBlur(img, (5,5), 0)  
  
cv2_imshow(blur)  
  
cv2.waitKey(0)  
  
cv2.destroyAllWindows()
```

OUTPUT



EXPERIMENT-3

CODE

```
import cv2

import matplotlib.pyplot as plt

img = cv2.imread('/content/Screenshot 2026-05-06 061946.png')

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

edges = cv2.Canny(gray, 100, 200)

plt.subplot(1,2,1)

plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))

plt.title("Original Image")

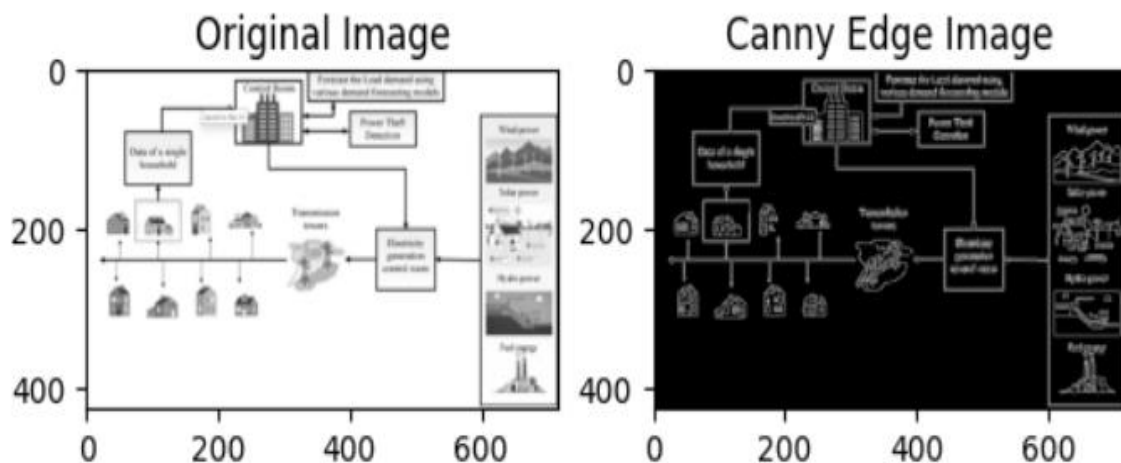
plt.subplot(1,2,2)

plt.imshow(edges, cmap='gray')

plt.title("Canny Edge Image")

plt.show()
```

OUTPUT

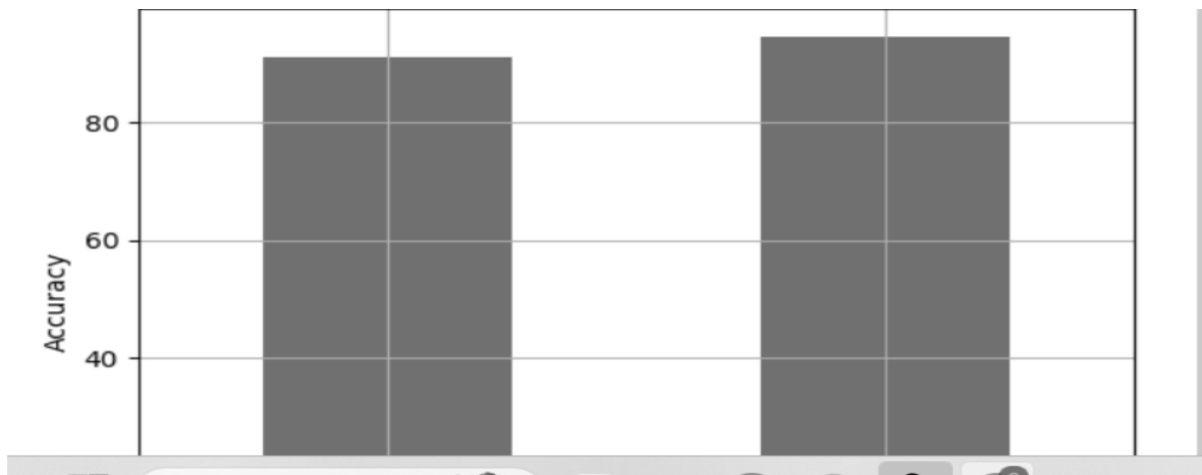


EXPERIMENT-4

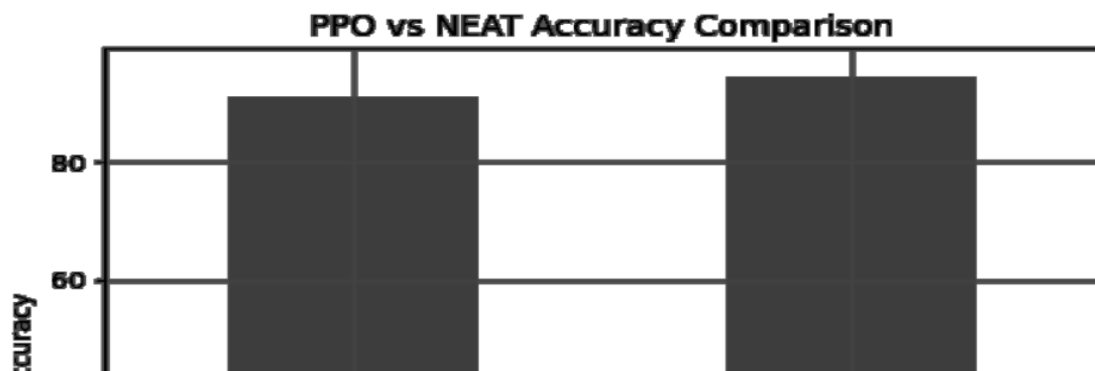
CODE

```
import cv2  
  
from google.colab.patches import cv2_imshow  
  
img = cv2.imread('/content/Screenshot 2026-04-24 213806.png', 0)  
equalized = cv2.equalizeHist(img)  
cv2_imshow(img)  
cv2_imshow(equalized)
```

OUTPUT



Standard Deviation:
Algorithm
NEAT 0.386907
PPO 0.451919
Name: Accuracy, dtype: float64



EXPERIMENT-5

CODE

```
import cv2

import matplotlib.pyplot as plt

img = cv2.imread('/content/Screenshot 2026-05-06 061946.png')

colors = ('b', 'g', 'r')

for i, col in enumerate(colors):

    hist = cv2.calcHist([img], [i], None, [256], [0,256])

    plt.plot(hist, color=col)

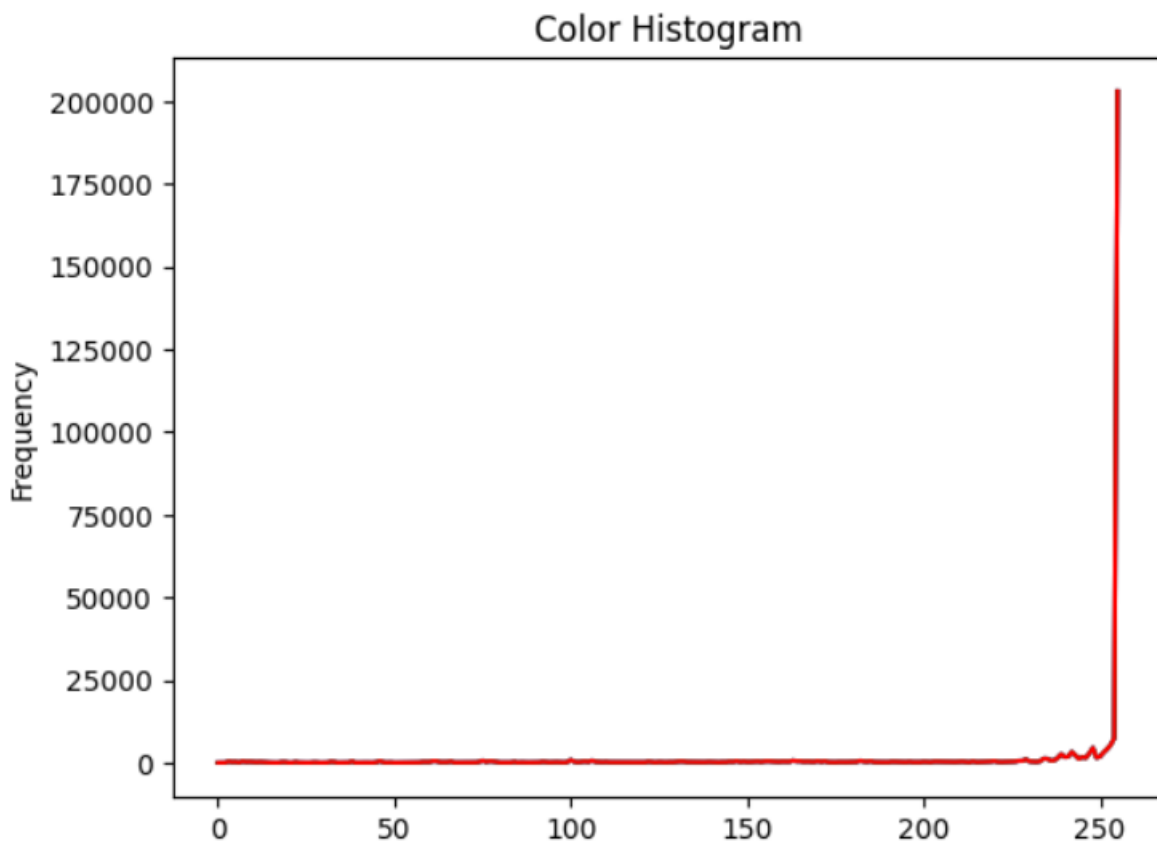
plt.title("Color Histogram")

plt.xlabel("Pixel Value")

plt.ylabel("Frequency")

plt.show()
```

OUTPUT



EXPERIMENT-6

CODE

```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow
img = cv2.imread('/content/Mahadev ♡.jpg')
kernel = np.ones((5,5), np.uint8)
erode = cv2.erode(img, kernel, iterations=1)
cv2_imshow(erode)
```

OUTPUT



EXPERIMENT-7

CODE

```
import cv2  
  
from google.colab.patches import cv2_imshow  
  
cap = cv2.VideoCapture('/content/IHEARTGOT.gif')  
  
while cap.isOpened():  
    ret, frame = cap.read()  
    if not ret:  
        break  
    cv2_imshow(frame)  
  
cap.release()
```

OUTPUT



EXPERIMENT-8

CODE

```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow
img = cv2.imread('/content/Mahadev ♡.jpg')
kernel = np.ones((5,5), np.uint8)
dilate = cv2.dilate(img, kernel, iterations=1)
cv2_imshow(dilate)
```

OUTPUT



EXPERIMENT-9

CODE

```
import cv2

from google.colab.patches import cv2_imshow

img = cv2.imread('/content/Screenshot 2026-04-24 213806.png')

big = cv2.resize(img, None, fx=2, fy=2)

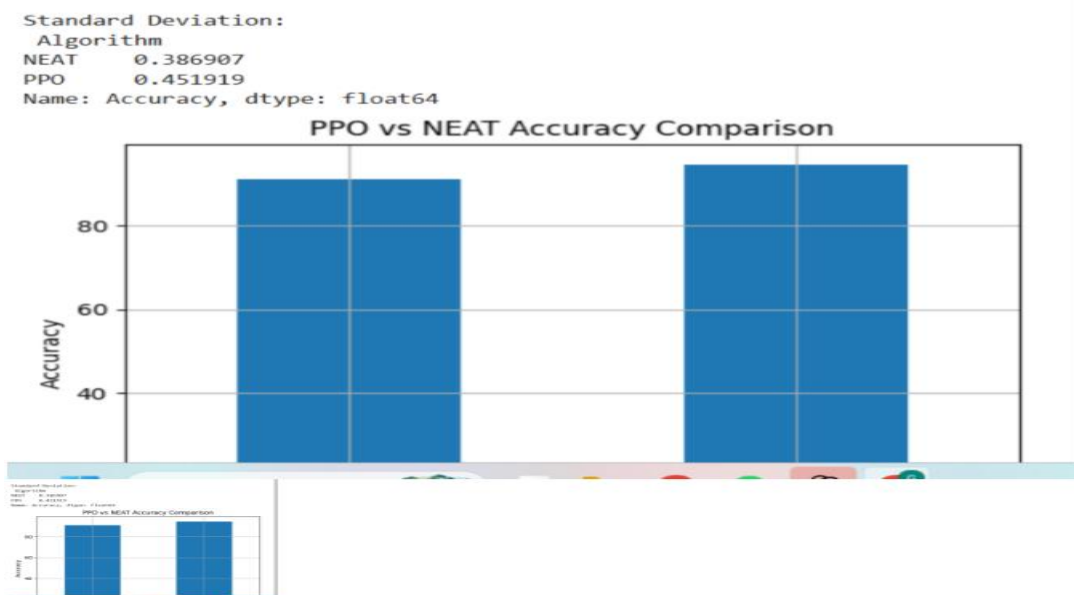
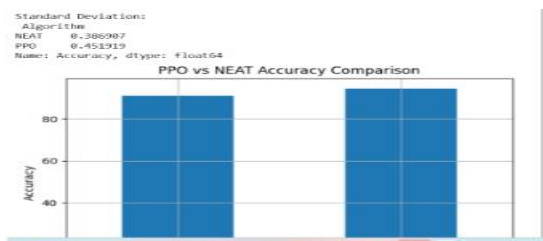
small = cv2.resize(img, None, fx=0.5, fy=0.5)

cv2_imshow(img)

cv2_imshow(big)

cv2_imshow(small)
```

OUTPUT



EXPERIMENT-10

CODE

```
import cv2  
  
from google.colab.patches import cv2_imshow  
  
img = cv2.imread('/content/Screenshot 2026-04-24 213806.png')  
rotate = cv2.rotate(img, cv2.ROTATE_90_CLOCKWISE)  
cv2_imshow(rotate)
```

OUTPUT

